💼 Project Title:

Customer Behavior and Sales Insights Dashboard

📌 Objective:

The goal of this project is to analyze customer behavior and sales trends using SQL and Power BI. The insights help identify high-value customers, purchasing patterns, and marketing opportunities, supporting data-driven decision-making.

📄 Dataset Overview:

- **Source:** [UCI Online Retail Dataset or Kaggle E-Commerce Data]
- **Size:** ~500K rows
- **Fields Used:**
    - Invoice Number
    - Customer ID
    - Country
    - Product Description
    - Quantity
    - Unit Price
    - Invoice Date
    - Total Revenue (calculated)

🛠 Tools Used:

- **SQL (PostgreSQL):** Data cleaning, filtering, and transformation
- **Power BI:** Data modeling, DAX measures, and dashboard creation
- **Excel/CSV:** Initial data format
- **pgAdmin :** Query interface

🔧 Steps Followed:

1. Data Preparation (SQL)

- Imported CSV into PostgreSQL using pgAdmin.
- Cleaned missing values, filtered null Customer IDs and negative quantities.
- Created a total_amount column using: total_amount = quantity * unit_price

2. Customer Segmentation with RFM (SQL)

- Used SQL queries to calculate:
    - **Recency:** Days since last purchase
    - **Frequency:** Count of unique invoices
    - **Monetary:** Total revenue per customer
- Scored customers based on RFM and grouped them into:
    - At risk
    - Loyal
    - Best customer
    - Other

3. Data Import to Power BI

- Connected PostgreSQL to Power BI via **PostgreSQL connector**
- Imported cleaned and aggregated tables/views
- Built a data model with proper relationships

4. DAX Calculations in Power BI

Key DAX measures and columns created:

- Total Orders = COUNTROWS(DISTINCT('public clean_sales'[invoiceno]))
- Total Revenue = SUM('public clean_sales'[totalamount])
- AOV = [Total Revenue]/[Total Orders]
- *Repeat Purchase Rate (%) =*

  VAR TotalCustomers = DISTINCTCOUNT('public customer_rfm'[customerid])

  VAR RepeatCustomers = CALCULATE(DISTINCTCOUNT('public customer_rfm'[customerid]),FILTER(VALUES('public customer_rfm'[customerid]),CALCULATE(MAX('public customer_rfm'[frequency])) > 1))RETURN

  DIVIDE(RepeatCustomers, TotalCustomers, 0) *100

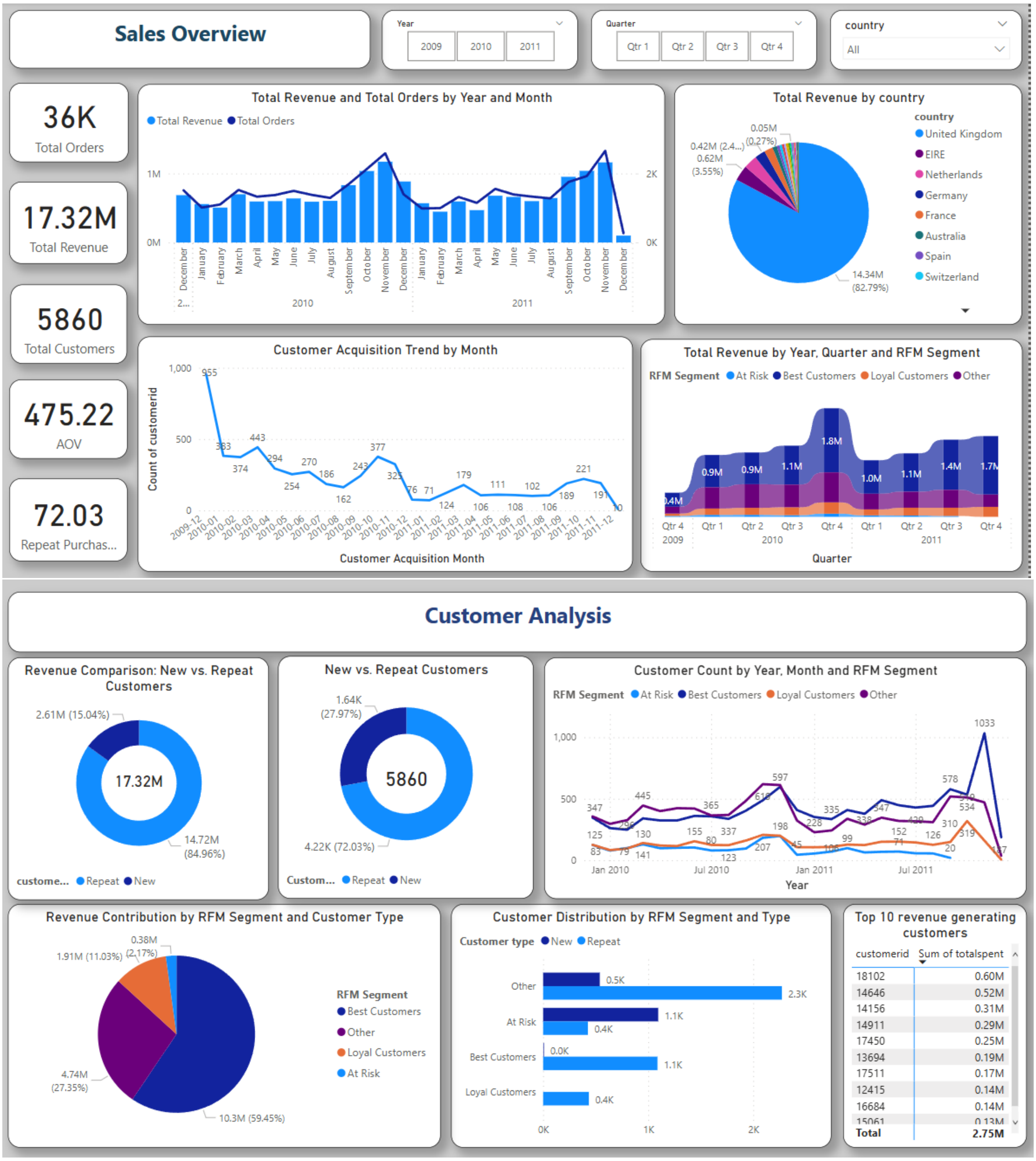
- *First Purchase Date =*

  CALCULATE(MIN('public clean_sales'[invoicedate]),
  ALLEXCEPT('public clean_sales' ,'public clean_sales'[customerid]))

- *Customer Acquisition Month =*
  FORMAT([First Purchase Date], "YYYY-MM")

- *Retention Flag =*
  IF ('public clean_sales'[invoicedate]> 'public clean_sales'[First Purchase Date],
  CALCULATE(1, ALLEXCEPT('public clean_sales', 'public clean_sales'[customerid])), 0

5. Power BI Dashboard Design



🔍 Key Insights Identified:

- Repeat customers contribute significantly to total revenue (around 85%).
- UK is the dominant market (83% revenue), followed by Germany and France
- Seasonal sales peaks around **November–December**.
- Retention rate of customers is around 72% (repeat customers).
- Customer Acquisition is seen to be *declining* over a period of time. Marketing team should focus on launching some customer acquisition campaigns.

📈 Impact / Value:

- Demonstrated how combining SQL with Power BI enables end-to-end analytics
- Showcased advanced DAX and dashboarding skills
- Positioned to support marketing and retention strategies with actionable insights